

White Paper

The Future of Pharmacovigilance: Integrating AI with Drug Safety Systems

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Introduction

The use of artificial intelligence (AI) is rapidly proliferating across industries, and the pharmaceutical space is no exception. Activating AI across every stage of the drug lifecycle can yield major benefits in terms of speed, efficiency, compliance, and patient outcomes. However, the relative newness of AI's widespread use gives many across the healthcare industry pause, particularly when it comes to ensuring compliance with a rapidly evolving regulatory landscape.

For pharmacovigilance (PV) teams, AI may be both appealing and daunting. On one hand, AI can be a major asset in terms of helping experts handle increasing case volume; on the other, implementing AI in PV requires strict tracking and documentation to ensure continued compliance with shifting guidelines. Regulators like the U.S. FDA have demonstrated an ongoing effort to educate themselves and provide guidelines for leveraging AI compliant with regulations, which is reflected in several initiatives by the agency to provide clarity and direction to stakeholders, including those in the PV space.

To successfully integrate AI into PV systems, an organization must align its people, technology, and processes to build streamlined workflows supported by a robust governance structure. If done successfully, these efforts will yield significant efficiencies across PV workflows.

Activating AI across every stage of the drug lifecycle can yield major benefits.



The current regulatory landscape

Despite a climate of uncertainty, the FDA and other regulators around the world have been active in terms of regulations and guidance around AI for nearly a decade (Figure 1). In 2016, the FDA published a paper that highlighted how it planned to review regulatory submissions with AI components. From 2016 through 2018, the FDA conducted heavy research on how to activate AI across different stages of the drug development lifecycle and throughout lifecycle management, particularly in the drug safety domain. And in the last four years, the FDA has organized internally and collaborated with other regulatory agencies to put out discussion papers and guidance.

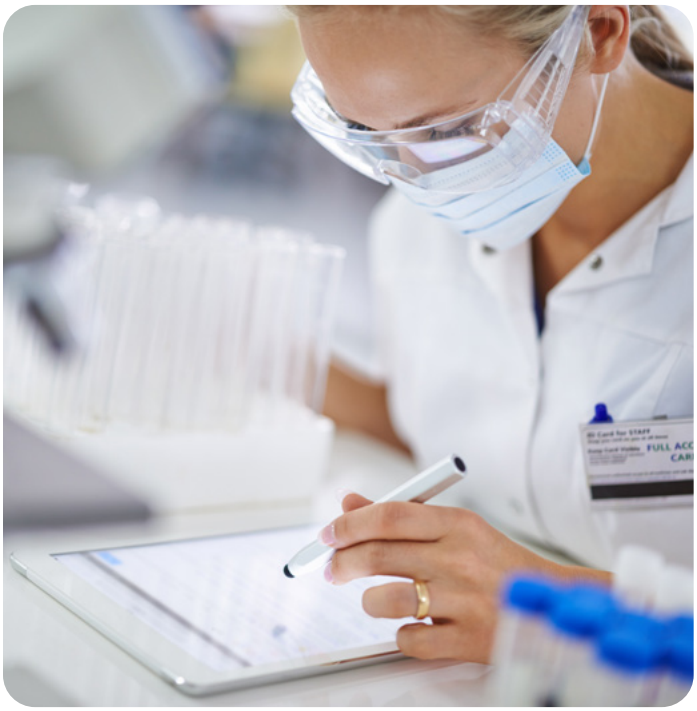
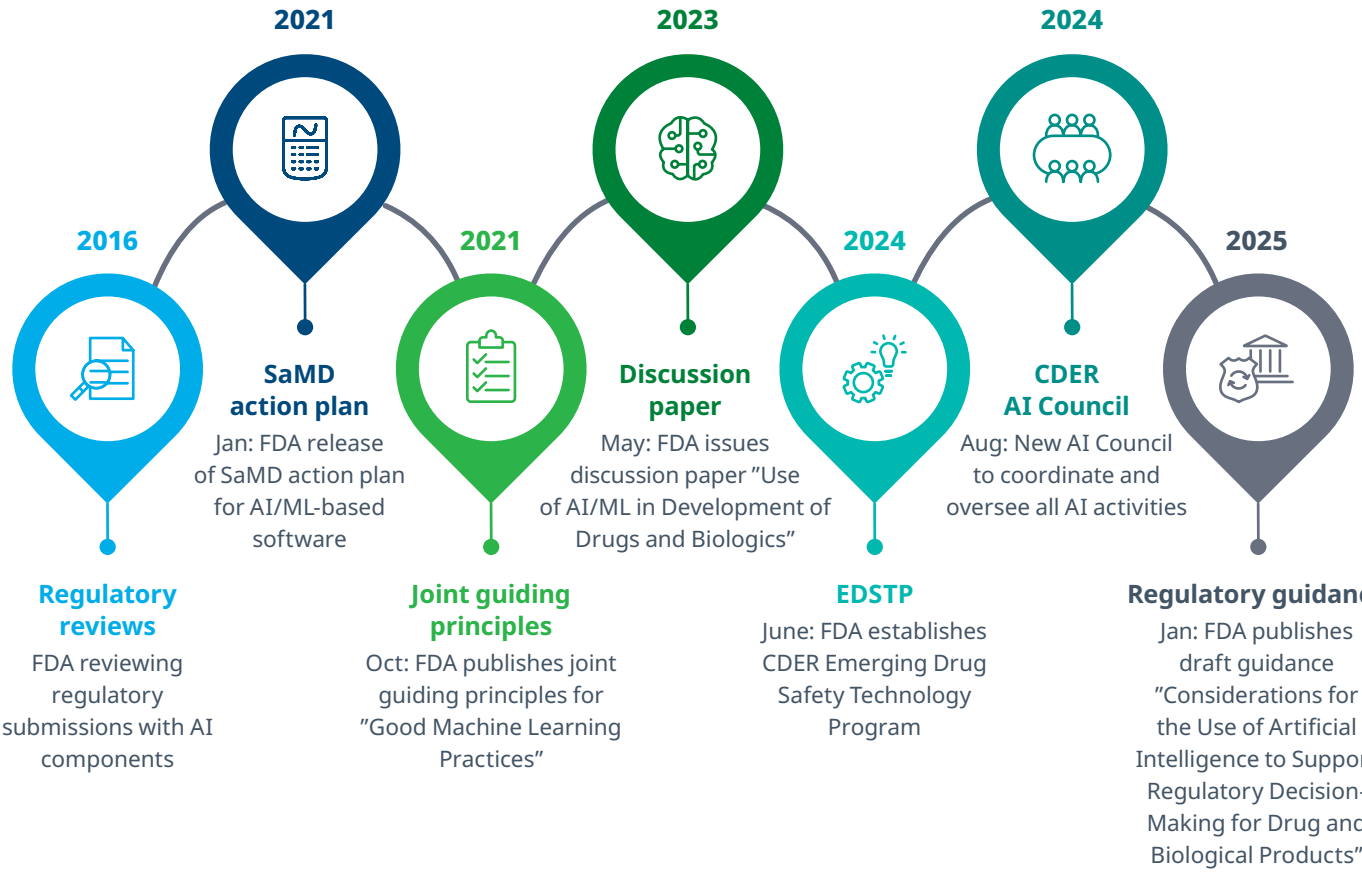


Figure 1: Timeline of FDA responses to AI





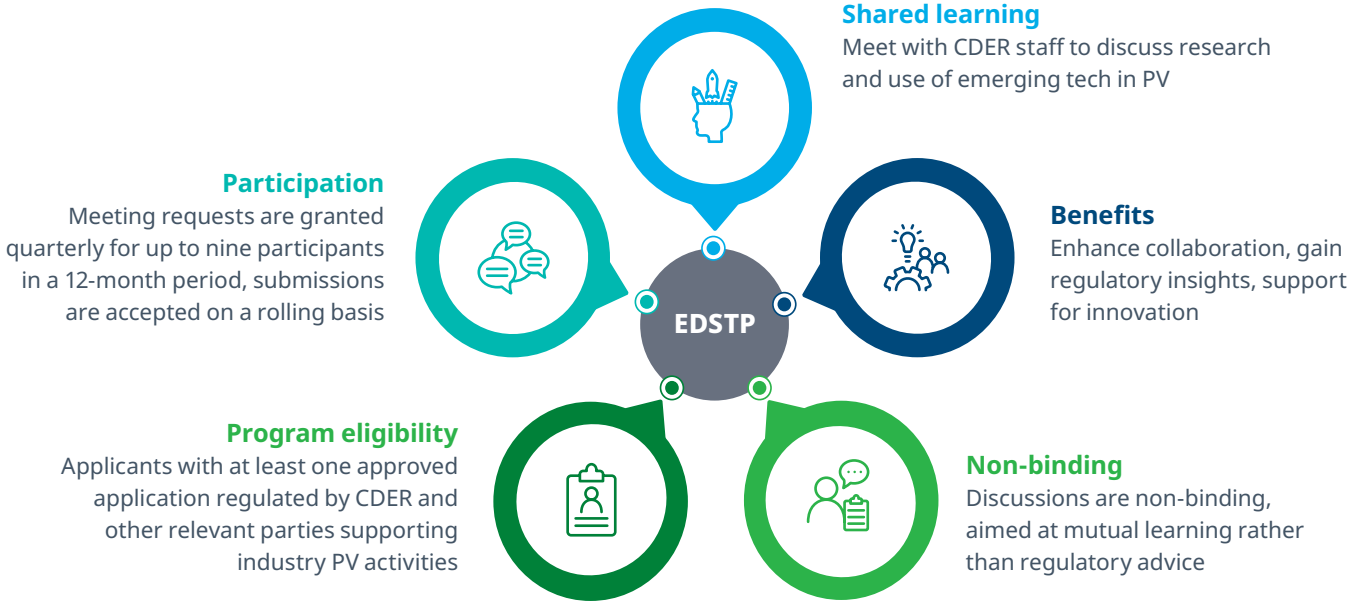
Fast forward to today, and the FDA continues to explore how drug manufacturers can leverage this technology in development and manufacturing. In January 2025, the agency published the draft guidance "Considerations for the Use of Artificial Intelligence to Support Regulatory Decision-Making for Drug and Biological Products," which aligns closely with the discussion paper they released in 2023. The 2025 guidance focuses on AI usage across all stages of the lifecycle for both small molecule drugs and biologics, demonstrating the FDA's willingness to not only embrace AI but to guide sponsors through its usage in a collaborative fashion. It specifically addresses how to use AI to support regulatory decision-making, including in the context of PV, outlining a risk-based approach to define an AI model to support credibility and accuracy.

The FDA's Center for Drug Evaluation and Research (CDER) recently launched the Emerging Drug Safety Technology Program (EDSTP), which intends to:

- Be a central point of discussion on the use of AI and other emerging technologies in PV.
- Manage and transfer knowledge specific to the context of AI use in PV.
- Better understand the use of AI and other emerging technologies in PV to inform regulatory approaches.

The FDA continues to explore how drug manufacturers can leverage this technology in development and manufacturing.

Figure 2: A breakdown of EDSTP



One of the key components of EDSTP is to provide sponsors, upon request, with the opportunity to meet with the FDA to discuss how AI is being used in their processes and to gain regulatory insights into using AI in the PV domain. The EDSTP meeting aims to create an opportunity for shared learning (Figure 2). These discussions are non-binding and open to applicants with at least one approved application regulated by CDER, as well as CROs and other relevant parties. Meeting requests are granted quarterly for up to nine participants in a 12-month period and submissions are accepted on a rolling basis.

These efforts by the FDA are all positive signs towards their support of the use of AI in PV practices. Though the space continues to evolve, regulators are striving to provide opportunities for collaboration to foster innovation in PV workflows.



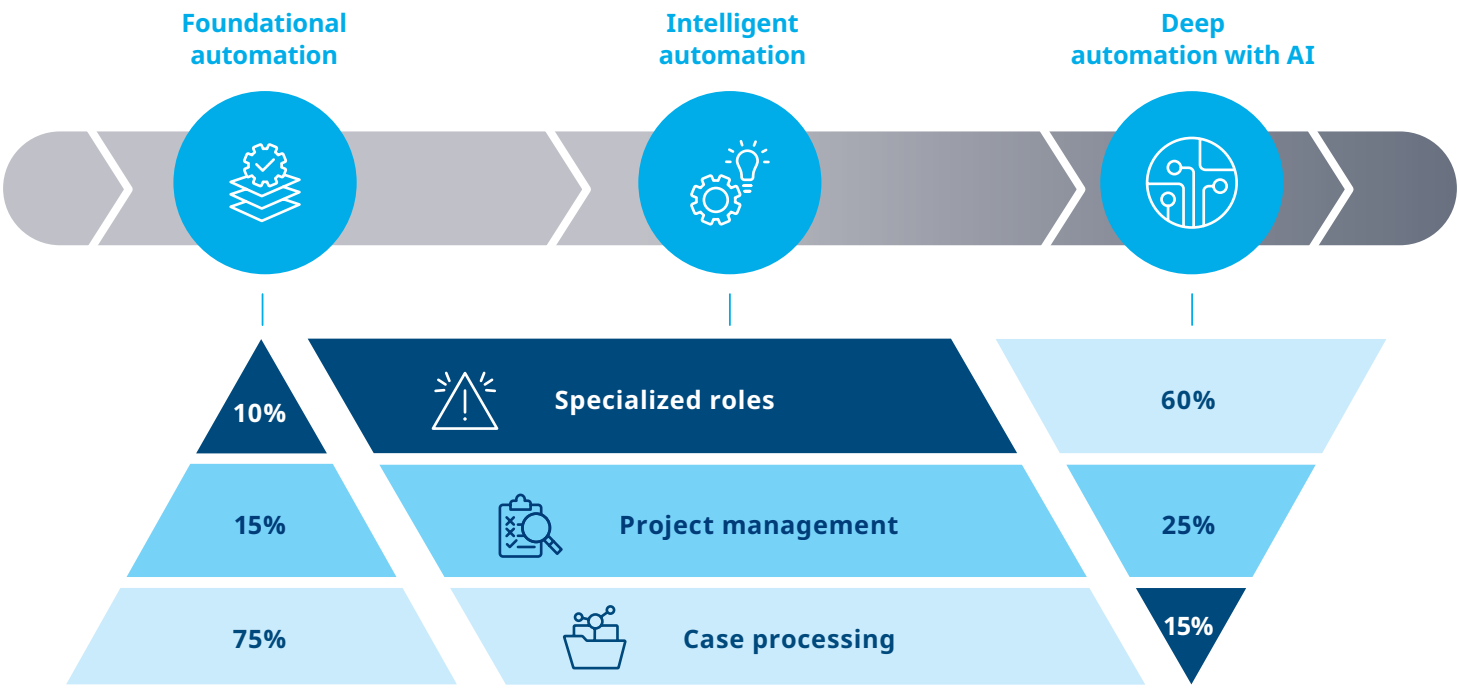
The next generation of PV

PV teams are navigating unprecedented levels of structured and unstructured data, which will continue to increase in complexity and volume from a variety of sources, including clinical trials, medical information calls, insurance claims, social media, and literature. To better accommodate this diversity of data, technology, including agentic AI and advanced analytics, must be integrated to ensure efficient processing and minimal handoffs. In terms of case processing, which includes data entry, medical review, quality management, and submissions, the key question is when and where to add technology and how to guarantee that subject matter experts (SMEs) are able to successfully leverage it to improve processes.

In the years ahead, PV practices will be impacted by any guidance that comes out from regulators, changes in technology, and the roles and skillsets of the workforce. To successfully navigate this next era of change in PV, processes must adapt to regulatory changes and new technology, which requires robust SME knowledge. At IQVIA, we are focused on developing Smart-Touch PV, which emphasizes bringing technology in at the right time to allow for process improvements while ensuring that PV SMEs can make a seamless transition to these changes.

Historically, the bulk of a PV team's resources have gone to case processing, followed by project management, with the least amount of resources going to specialized roles (Figure 3). However, with AI automation, that distribution will essentially flip. As a result, there will be fewer people in the case processing space, and more spending their time in project management and specialized roles. In specialized roles, PV team members will need to demonstrate industry and process knowledge as well as a tech savviness that allows them to interpret data, make strategic decisions, and justify said decisions. Given these responsibilities, specialized roles will be vital to the success of technology in PV workflows.

Figure 3: The evolving PV workload with AI assistance





As data becomes more complex and higher in volume, one of the critical areas of technology investment is literature surveillance, which helps enable the rapid triage and identification of documents relevant for a drug or medical device. Another is ensuring efficient and intelligent processing of data generated from medical information contact centers. Safety systems also present an opportunity to embed AI. The IQVIA Vigilance Platform constitutes a comprehensive safety system that encompasses modules designed for safety detection, data collection, case processing, periodic submissions, and signal detection, all enhanced by artificial intelligence to facilitate smart touch.

To apply AI in the most effective and beneficial way, it is important to consider return on investment for the functions that it is assisting and the impact on an overall portfolio. Teams must identify the most appropriate use cases and determine the change management deliverables that must be put in place as well as any relevant regulatory, legal, IP, and data sensitivity considerations. Ultimately, the most essential consideration is continued patient safety. As AI technology is used in new processes, keeping expert personnel in the loop ensures a deliverable has been vetted and approved by an SME for safety and compliance before it is released.

The IQVIA Vigilance Platform constitutes a comprehensive safety system that encompasses modules designed for safety detection, data collection, case processing, periodic submissions, and signal detection.

The IQVIA approach

The IQVIA team is focused on leveraging AI-enabled technologies throughout the drug development and manufacturing lifecycle, using insights from each stage to inform the next. With AI-driven workflows, we are able to provide customers with end-to-end services that are designed with their most critical challenges in mind. The IQVIA Healthcare-grade AI™ system was built to meet the unique demands of healthcare by expertly leveraging the right data, technology, and expertise side-by-side. Healthcare-grade AI includes a "human-in-the-loop" as part of the model, allowing AI to be tracked and monitored within a governance structure to verify that it is consistently reliable (Figure 4). Over time, with proven success and reliability, the option to turn off the human-in-the-loop component may arise.

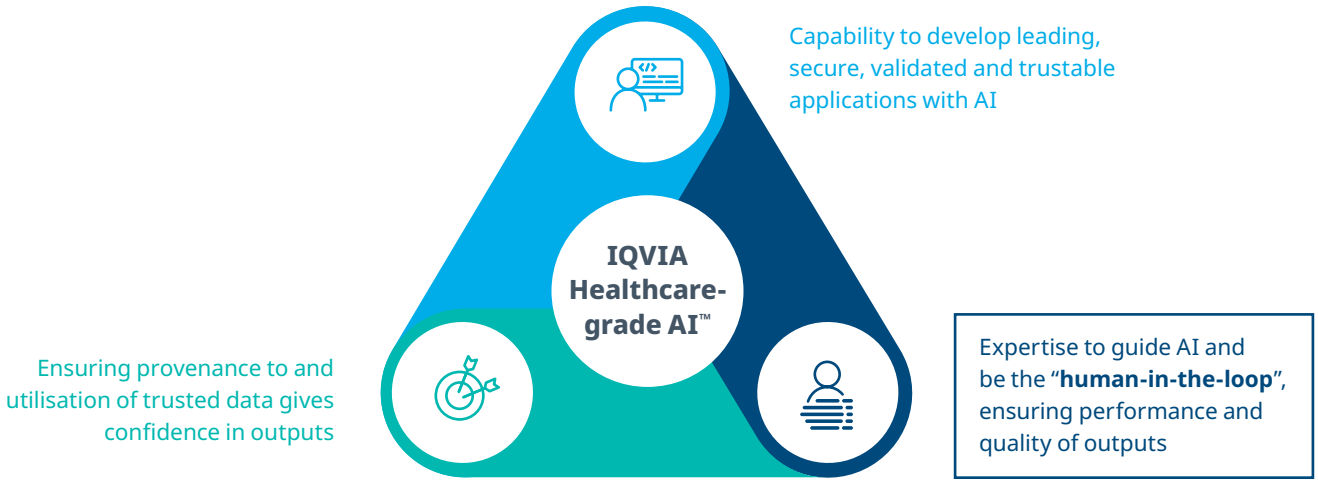
The IQVIA Vigilance Platform leverages several different modules that are supported by AI, many of which will be going live within the next year. One example includes the use of AI for in-operation data analysis — one of the most valuable assets that AI provides. In the background of a medical information call, AI can conduct voice analysis, transcription, and identify adverse events. Sentiment analysis is incorporated to provide greater insight to the safety team when a project is handed off.

By conducting this analysis in the background, AI enables fast, efficient, and high-quality workflows. GenAI language processing is applied to social media to scan vast pools of unstructured data for adverse events or for text extraction in long documents. AI-enabled optical character recognition (OCR) is used in form extraction to pull data from structured documents and conduct mapping to structured fields.

In terms of IQVIA's roadmap vision for AI in PV, AI can be used to create narratives for periodic reports, signal evaluations, and medical assessments, providing the right data to a medic so that they are able to make an informed decision about whether a previous case is relevant to the case at hand. Over time, the AI system could begin to triage and prioritize those cases, providing insights and, thus, solutions for the challenges of not only today but tomorrow.

Ideally, regulators want to see how companies are evolving their PV systems and navigating challenges to use AI responsibly, rather than activating AI without ensuring that it is safely fulfilling its designated role. It is vital to gather credible evidence about the combined PV and AI system to demonstrate to regulators that a plan was designed and delivered upon. With the right governance structure, AI technology will be used to significantly improve PV processes.

Figure 4: The critical components of IQVIA Healthcare-grade AI





Looking ahead

The use of AI in PV can enhance processes such as data analysis, signal detection, and case management by increasing efficiency, accuracy, and the ability to manage large datasets. To successfully leverage this technology, sponsors must deftly navigate the evolving regulatory landscape, taking opportunities where possible to collaborate with regulatory authorities and share insights. Opting to work with an experienced provider of AI-driven PV solutions will help your team use this technology to its maximum benefit.

About IQVIA

IQVIA (NYSE:IQV) is a leading global provider of clinical research services, commercial insights and healthcare intelligence to the life sciences and healthcare industries. IQVIA's portfolio of solutions are powered by IQVIA Connected Intelligence™ to deliver actionable insights and accelerate innovations. With approximately 88,000 employees in over 100 countries, IQVIA is dedicated to accelerating the development and commercialization of innovative medical treatments to help improve patient outcomes and population health worldwide. Learn more at www.iqvia.com.

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About the authors



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Archana Hegde joined IQVIA in 2015 and has been in various leadership roles implementing technology solutions to support critical business processes. In her current role, she provides senior leadership and oversight to the Safety Tech-Service Integrated Offering delivery, combining IQVIA's expertise from a technology and service perspective while focusing on accuracy, efficiency, compliance, standardization, and automation.



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Updesh is responsible for developing the overarching strategy regarding AI and machine learning as it relates to safety and pharmacovigilance. He has over 25 years of knowledge and experience in the management, development, implementation, and operation of processes and systems within the life sciences and other industries.

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